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MILITARY SPECIFICATION

CARTRIDGE, BALL, CALIBER .22, LONG RIFLE (COMMERCIAL)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers cartridges intended for use in caliber .22 small arms weapons.

1.2 Classification. The cartridges shall be of three grades:

Match St
Standard velocity
High velocity

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, form a part of this specification.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

42–23–2 — Rifle, U. S. Caliber .22, M2.

42–29–1 — Rifle, Winchester, Caliber .22, Model 75, Target; or Rifle, Winchester, Caliber .22, M52.

42–30–1 — Rifle, Remington, Caliber .22, Match Master, Model 513T; or Rifle, Remington, Caliber .22, M40X.

49–15–1 — Gage, Pressure, Caliber .22, M52.

A47753 — Cylinder, Pressure.

B7553931 — Cartridge, Caliber .22, Long Rifle (Commercial).

D6535485 — Trainer, Machine Gun, Caliber .22, M3.

D6535495 — Trainer, Machine Gun, Caliber .22, M4

FRANKFORD ARSENAL

FD 14912 — Calibers .22, .30, 30 Carbine, 7.62 mm, Caliber .38, .45 Jaws, Jaw Holder, Case Holder, etc. (for Machine, Bullet Pull Test).

FSC 1305

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PUBLICATIONS

ORDNANCE CORPS

ORD–SIP–S315 — Visual Inspection Standards for Small Arms Ammunition.

ORD–M608–PM, Vol. III — Manual of Test Methods for Small Arms Ammunition.

List of Inspection Gages — List of Inspection Gages for Cartridge, Ball, Cal. .22, Long Rifle (Commercial)—7553931.

(Copies of specifications, standards, drawings and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other Documents. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

INTERSTATE COMMERCE COMMISSION

49 CFR 71–78 — Interstate Commerce Commission Rules and Regulations for Transportation of Explosives and Other Dangerous Articles.

(The ICC regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publications should cite "49 CFR 71–78 (Rev. 1950").

3. REQUIREMENTS

3.1 Design and construction. The cartridge shall comply with all requirements specified on drawing B7553931.

3.1.1 *Construction.* All cartridges shall be manufactured in accordance with a procedure approved by the contracting officer. No deviation from the approved manufacturing procedure shall be made without prior approval of the contracting officer.

3.1.2 *Material.* Material shall conform to requirements of applicable specifications and drawings. The contractor shall furnish the Government inspector copies of chemical, physical and micrographic tests made on material used in the manufacture of the cartridges.

3.1.3 *Components.* All components shall have been inspected and tested in accordance with the applicable drawings and specifications, prior to assembly.

3.1.3.1 *Case.* The metal shall be brass.

3.1.3.2 *Bullet.* The metal shall consist of lead or a lead alloy.

3.1.3.2.1 *Bullet coating.* Each bullet shall be coated with a moisture proof material having lubricating properties.

3.1.3.3 *Propellant.* The propellant shall be smokeless type.

3.1.3.4 *Primer mixture.* The mixture shall be noncorrosive.

3.1.4 The contractor shall maintain a system of processing which will insure that the process average quality with respect to all requirements is equal to or better than the acceptable quality levels specified herein.

3.2 Bullet extraction. The force necessary to extract any one bullet from its case shall not be less than 20 pounds, and the average force necessary to extract bullets from cases shall not be less than 30 pounds. Match grade cartridges are exempt from this test.

3.3 Velocity. The basic velocity at 15 feet shall be as follows:

Grade	*ft/sec*
Match	[1]
Standard velocity	1135
High velocity	1315

[1] To be recorded.

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The average velocity shall not vary from the basic velocity by more than 25 feet per second.

3.4 **Pressure.** The average pressure shall not exceed 24,000 pounds per square inch.

3.5 **Accuracy firing.** The average extreme spread, center to center, of all targets at 100 yds shall not exceed the following limits:

Grade	*inches*
Match	1.5
Standard velocity	2.2
High velocity	2.5

3.6 **Function and casualty firing.** Total defects resulting from firings in this test and all other firings shall not exceed the percentages specified in table III. Match grade ammunition shall be fired in rifles only.

3.7 **Noncorrosiveness.** No rust shall be detected upon visual examination of the bore of the test barrel upon completion of the noncorrosiveness test.

3.8 **Heat.**

3.8.1 *Coating.* There shall be no hardening, flaking or melting of the bullet coating when subjected to the heat test.

3.8.2 *Accuracy.* Targets of all cartridges fired as part of the heat test shall have an average extreme spread not greater than 2.0 inches for Match grade, 2.7 inches for Standard Velocity grade, and 3.0 inches for High velocity grade.

3.8.3 *Velocity.* The average velocity of the heated cartridges shall be not more than 50 feet per second below the recorded velocity of the lot of cartridges. Pressures shall be recorded for information.

3.9 **Workmanship.**

3.9.1 Ammunition shall be free of visual and gaging defects listed in Table I, to the extent of the acceptable quality levels specified in paragraph 4.3.4. All parts shall be free of chips, dirt, grease, rust and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

3.9.2 Reworking, repairing, remachining, reassembly or reprocessing of cartridges, components, or component assemblies which fail to conform to applicable drawings and specifications after passing through the original process or which may be salvaged from breakdown operations, shall not be presented for acceptance without prior approval of the contracting officer. Such cartridges, components or component assemblies when presented shall be free of defects. The contracting officer shall require segregation and separate presentation to the Government inspector of such ammunition or ammunition containing such components or component assemblies.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Lot.**

4.1.1 *Ammunition lot.* A lot shall consist of cartridges produced by one manufacturer in accordance with stabilized methods and techniques, the same drawing and drawing revision, same specification and same specification revision.

4.1.2 *Inspection lot.* The contractor may submit a lot for visual inspection and gaging by inspection lots. Unless otherwise specified, inspection, gaging, and testing may not be started before each lot is assembled and packed, except that the packing cases need not be closed until the lot is accepted.

4.1.3 *Lot identification.* Each lot of ammunition shall be identified with a lot number. No lot number shall be assigned to more than one lot of ammunition containing the same type, caliber and model of cartridge.

4.1.4 *Test samples.* Cartridges expended

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from the lot for required tests for acceptance shall be considered as deliveries under the contract if the lot is subsequently accepted. When a lot is retested at the request of the contractor and for the benefit of the contractor, samples expended in the retest shall not be considered as deliveries under the contract.

4.2 **Sampling.** Sampling of an ammunition lot for inspection and gaging shall be selected by the Government inspector in accordance with inspection level II of Standard MIL–STD–105.

4.3 **Inspection.** At the discretion of the inspection agency, any inspection specified herein for accomplishment by the Government will be interpreted to mean inspection to be accomplished either by or under the supervision of the Government, with the final decision as to acceptance or rejection in either case resting with the Government.

4.3.1 *Place.* Unless otherwise specified, inspection, gaging, and testing may be done at the plant of the contractor or point of delivery, or both.

4.3.1.1 *Inspection equipment.* All inspection gages and testing equipment required by the Government inspector will be furnished by the Government and shall be under the control of the Government inspector. Such equipment may be on a loan basis to the contractor for his required tests.

4.3.2 Each cartridge of the acceptance sample shall be inspected and gaged for all defects listed in table I. Any defects discovered by the Government inspector not listed in table I shall be recorded on the acceptance report and brought to the attention of the contracting officer.

4.3.3 The cartridges presented for inspection, gaging, and test shall be under the complete control of the Government inspector. The contractor shall furnish adequate and proper storage facilities and working areas to the Government inspector.

4.3.4 *Acceptable quality level.* The acceptable quality level with respect to visual and gaging defects shall be as follows:

Defects	*Percent*
Major	0.25
Minor	1.5

4.3.4.1 If the number of defective cartridges be equal to or greater than the applicable rejection number, the inspection lot shall be rejected subject to the provisions of paragraph 4.4.12.

4.3.5 *Classification of defects.* The visual and gaging defects shall be classified as specified in table I. Gaging inspection shall be accomplished with the use of Ordnance certified gages listed on List of Inspection Gages For Cartridge, Ball, Cal. .22, Long Rifle (Commercial)—7553931.

TABLE I. *Defect classification*

No.	Defect	Major	Minor	Major or Minor
	Visual[1]			
	Cartridge			X
1	Corroded or stained			X
2	Crooked[2]			
	Case			
3	Dented or mashed[2]			X
4	Split	X		
5	Perforated	X		
6	Draw Scratch			X
7	Draw Rough	X		
8	Scratch		X	

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TABLE I. *Defect classification*—Continued

No.	Defect	Major	Minor	Major or Minor
9	Scaly Metal			X
10	Fold		X	
11	Wrinkle		X	
12	Bulge		X	
13	Defective Head[3]			X
14	Defective mouth		X	
	Bullet			
	Excessive lubricant coating		X	
15	No lubricant coating		X	
16	Dent		X	
17	Scratch		X	
18	Gash	X		
19	Upset point		X	
20	Flat point		X	
21	Jam	X		
	Gaging			
	Cartridge			
	Failure to chamber	X		

[1] Refer to ORD–SIP–S315 for definite classification of defect numbers 1–21.
[2] Classified as major if cartridge fails to chamber in a chamber gage, otherwise classified as minor.
[3] Classified as major if defect is such as to cause misfire, or rupture.

4.4 Tests. Various acceptance tests shall be as specified in table II. These cartridges shall be selected from the composite inspection samples which have passed the visual inspection.

TABLE II. *Test Samples*

Test	Number of Cartridges
Bullet extraction	50
Velocity } Pressure } [1]	24
Accuracy (10 warmer and 100 test cartridges in each of two weapons)	220
Noncorrosiveness	10
Heat test	
Accuracy (10 warmer and 100 test cartridges in each of two weapons)	220
Velocity	24
Function and casualty	
Rifle, U. S., Caliber .22, M2	150
Rifle, Winchester, Caliber .22, Model 75, or Rifle, Winchester, Caliber .22, Model 52	150
Rifle, Remington, Caliber .22, Model 513T, or Rifle, Remington, Caliber .22, Model 40X	150
Pistol, Automatic, Colt, Caliber .22, Woodsman	150
Pistol, Automatic, Hi-Standard, Caliber .22, H-D	150
Revolver, Colt, Caliber .22, Officers, Model Target	150
Revolver, Harrington & Richardson, Caliber .22 Sportsman Model, No. 999	150
Trainer, Machine Gun, Caliber .22, M3 or M4	300

[1] Velocities and pressures are measured simultaneously.
Note. The sample sizes shown above shall be the minimum quantity to be tested. This quantity may be increased at the discretion of the Government inspector.

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4.4.1 *Bullet extraction.* Each cartridge of the test sample shall be tested in an approved type tensile test unit, capable of registering within a minimum range of 0 to 100 pounds tension and using jaws conforming to Drawing FD 14912 or equivalent. The rate of travel of the test head shall be controlled within 3 to 6 inches per minute, inclusive. The extractive force required to separate the bullet from each cartridge case shall be recorded.

4.4.2 *Ballistic tests.* All ballistic tests shall be fired in accordance with procedures specified in publication ORD–M608–PM, Vol. III.

4.4.3 *Pressure firing.* Cartridges shall be fired in a weapon conforming to Drawing 49-15-1 held in an approved test. The velocity shall be taken simultaneously with pressure. The first, second, thirteenth, and fourteenth shots are to be discarded as they are intended to seat the pressure piston with the obturating cup. The obturating cups are to be changed after each twelve shots. Red wax, or equivalent, is to be used in the obturating cup. Pressure cylinders shall conform to Drawing A47753.

4.4.4 *Velocity firing.* This test shall be fired simultaneously with pressure test.

4.4.5 *Accuracy firing.* One half of the test sample of cartridges shall be fired in each of Rifle, Cal. .22, Winchester, M52, Heavy Barrel Type and Rifle, Cal. .22, Remington, 40X. The contractor's procedure for firing the accuracy test may be used unless otherwise specified by the contracting officer. The Government inspector shall witness the firing and computation of accuracy.

4.4.6 *Noncorrosiveness.* Ten cartridges shall be fired in a Barrel, from any weapon shown in table II which, after firing and without cleaning the bore, shall be placed in a humidity chamber with circulating air, so that the axis of the bore is parallel to the path of air. The chamber is to be maintained at 115-120 degrees Fahrenheit and 75-85 percent relative humidity. After 48 hours exposure, the barrel shall be removed and the bore visually examined for rust. Several clean cloth patches shall be passed through the bore and examined for evidence of rust as distinguished from powder residue.

4.4.6.1 A noncorrosiveness test shall be performed for at least one ammunition lot from each cartridge type covered by a procurement contract. Provided no change in powder, bullet lubricant, or priming compositions have been made, additional noncorrosiveness test for the same cartridge type may be waived after the acceptance of the first ammunition lot at the discretion of the contracting officer.

4.4.7 *Heat test.* Three hundred cartridges (six boxes of 50 cartridges each) shall be selected. Four cartridges shall be selected from the middle of each box and stored at 70°F. at ambient humidity as a normal sample for velocity test. These cartridges shall be replaced in the boxes by others of the same type from the same ballistic sample. The 300 cartridges shall be subjected to a temperature of 115° to 120°F. with a relative humidity of 75 to 85 percent for a period of 48 hours, after which they shall be normalized by storing at 70°F. at ambient humidity for at least two hours. After normalization 4 cartridges shall be selected from each box and the 24 cartridges shall constitute the heat test velocity sample.

4.4.7.1 *Accuracy.* Two hundred cartridges shall be selected at random from the heat test sample and ten targets of ten shots each fired at a range of 100 yards, in each of two rifles, Winchester, Caliber .22, M52, Heavy Barrel Type, and Remington, Caliber .22 Model 40X, which have been qualified for use by test with ammunition of known accuracy performance, as provided in paragraph 4.4.5. Immediately prior to firing for record ten warming rounds shall be fired in each weapon.

4.4.7.2 *Velocity.* The 24 cartridges set aside as a normal sample shall be fired for velocity and pressure in a velocity and pressure bar

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rel, drawing 49–15-1, followed by the 24 cartridges set aside from the heat test sample. Obturating cups shall be changed after each twelve shots. The first, second, thirteenth, and fourteenth velocities and pressures from each series shall not be included for record as they are intended to seat the pressure piston with the obturating cup. Pressures shall be recorded for information only.

4.4.7.3 A heat test shall be performed for at least one ammunition lot from each cartridge type covered by a procurement contract. Provided no change in powder, bullet lubricant, or priming compositions have been made, additional heat tests for the same cartridge type may be waived after the acceptance of the first ammunition lot at the discretion of the contracting officer.

4.4.8 *Function and casualty.* In all tests, the weapon shall be at ambient temperature and cooled after each 100 rounds tested. In the machine gun, the cartridges shall be fired in bursts of 100 cartridges. In the rifles and pistols the cartridges shall be fired rapid fire with intervals of not more than one-half minute between magazines. In revolvers the cartridges shall be fired, one half at double action and one half at single action, rapid fire and cylinders reloaded as rapidly as feasible. Cartridges shall be fired from weapons listed in table II, except that Match grade ammunition shall be fired in rifles only.

4.4.8.1 *Firing defects.* Firing defects obtained in all weapons, except machine guns, shall not exceed the limits specified in table III. Firing defects encountered in machine gun firings shall be recorded for information only.

4.4.9 *Breeching space.* All breeching spaces shall be 0.042 to 0.050 inches inclusive and measured in accordance with Publication ORD–M608–PM, Vol. III.

4.4.10 *Defective equipment.* When any firing defect is found to have been caused by defective equipment, such firing defect shall not be counted against the ammunition but the equipment shall be corrected or replaced and the test repeated. This also applies in any retest of ammunition.

TABLE III. *Firing defects and limits permitted in a sample*

Defect	Percent permitted
Failure to chamber	0.15
Misfire	0.15
Bullet remaining in bore[1]	0.00
Pierced case	0.15
Ruptured case, complete[2]	0.07
Ruptured case, partial, within ⅛" of face of head.	0.07
Ruptured case, partial[3]	0.15
Split case, serious[4]	0.07
Split case, non-serious[5]	0.20
Cracked rim[6]	0.07
Failure to extract	0.15

[1] No retest permitted. Lot shall be rejected.
[2] A complete rupture is a circumferential separation whereby the case separates into 2 pieces.
[3] A partial rupture is a circumferential separation of less than 360° in magnitude.
[4] A serious split is one which is through the head or through the side-wall within ⅛ inch from the face of the head.
[5] A nonserious split is one which occurs in the body of the case.
[6] A cracked rim is a crack ⅛ inch long or less, a portion of which is under the firing pin indent.

4.4.11 *Firing defects.* In any ballistic test, except function and casualty machine gun firing, in which a misfire occurs or a bullet remains in a bore, an additional shot shall be fired. That particular test shall not be penalized but the acceptance sample shall be penalized for such defects in accordance with Table III.

4.4.11.1 *Unusual firing defects.* Any defects not covered by this specification that are found by the Government inspector during the acceptance test shall be recorded on the acceptance report and brought to the attention of the contracting officer. The contracting officer shall report the defects to the Chief of Ordnance along with comments of the manufacturer when defects indicate that the ammunition is unsuited for the purpose intended.

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4.4.12 *Retests and reinspection.*

4.4.12.1 *Visual and Gage.* At the discretion of the contracting officer, a lot of cartridges which has failed to meet acceptable visual and gage quality requirements upon first acceptance test and has been reinspected and retested by the contractor, may be resubmitted for a second acceptance test. No one lot of cartridges may be submitted by the contractor for acceptance which has been previously rejected twice by the Government.

4.4.12.2 *Firing retest.* If the sample fails one or more of the firing tests, the retest shall be limited to the failing test or tests, but, if in the retest, other defects develop in excess of the percent permitted on original test, the lot shall be rejected. Retests for firing defects, except bullet remaining in bore, shall be made by firing double the sample size specified in table II for that phase, or phases, in which the failure occurred. If the percent permitted in table III is exceeded in the retest, the lot shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Level C.

5.2 Packing and Packaging. Unless otherwise specified in the contract or purchase order, cartridges shall be packed and packaged in commercial type containers as follows:

Paperboard box	*Carton*	*Packing box*
50 cartridges	10 paper board boxes	20 cartons

The total shall be 10,000 cartridges per box unit.

5.2.1 Unless otherwise specified in the contract or purchase order, the packaging material shall be of the best commercial quality, as furnished by the contractor.

5.3 Marking and labeling. Each paperboard box, carton and exterior shipping container shall be marked on the outside with the manufacturer's name or identification mark. The ammunition lot number shall be placed on each paperboard box and exterior shipping container. Marking of the lot number on the paperboard box may be either inside or outside at the manufacturer's option, but marking of the exterior container shall be on the outside. The Federal Stock Number, and Department of Defense Ammunition Code shall be placed on the exterior shipping container.

5.4 Delivery. A lot which has been officially inspected and accepted by the Government inspector shall be considered as delivered when the cartridges are packed, the ammunition packing sealed and marked as required and the packed ammunition has been accepted. After delivery has been accomplished as specified herein, the lot shall be either shipped or stored, in accordance with the terms of the contract as may be directed by the Government inspector.

5.4.1 When the contract provides for delivery f.o.b. cars at the place of manufacture, the ammunition shall be placed and properly secured in accordance with 49 CFR 71–78.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number and date of this specification.

6.2 The Government will furnish the following equipment for use in acceptance testing: all rifles, gages, pistols, and machine gun trainers, including all mounts, tools, gages, pressure cylinders with tarage table, and accessories indicated by drawings listed in section 2.

6.3 Provisions should be incorporated in each inquiry or invitation for bids as follows:

> "If the bidder desires to furnish a cartridge which in any way differs from the requirements of this specification, a statement to that effect shall be submitted with the bid. This

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statement shall describe fully the proposed difference. In addition, he shall submit full and sufficient evidence that such proposed cartridge is equal in all respects to the cartridges covered by this specification. The bidder shall submit samples of the proposed cartridge, at the request of the contracting officer, for such inspection and tests as are considered necessary to determine whether the proposed ammunition complies with the requirements of the specification".

6.4 Rifles will be verified in the following particulars before being issued for test purposes:

(1) Firing pin protrusion:
Rifle, U.S., M2–0.032 to 0.036 inch.
Other arms—according to commercial standards.

(2) Head space—0.042 to 0.050 inch. (Head space for accuracy rifles may be according to commercial practice).

(3) Correct shape of firing pin.

(4) Correct chamber and bore dimensions.

(5) Proper function of magazine.
Such rifles will not be weapons of exceptional quality but are to be standard stock. Pistols and revolvers are included in the same category.

6.5 Information in detail can be obtained on M3 and M4 machine gun trainers from the following publications of Springfield Armory:

Description of Trainer, Machine Gun, Calber .22, M3, Converted from Browning Machine Gun, Caliber .30, M1917 (or M1917A1).

Description of Trainer, Machine Gun, Caliber .22, M4, Converted from Browning Machine Gun, Caliber .30, M1919A4.

6.6 If during the gaging operation the lubricating grease on the bullet adheres to and builds up on the gage to prevent further gaging, the gage may be degreased prior to the cartridge gaging operation.

Notice. This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions is intended for use only in connection with procurement by the United States Government and shall not be used for any other purpose except when specifically authorized by the Chief of Ordnance.

Custodians:
Army—Ordnance Corps
Navy—Bureau of Ordnance
Air Force

Other interest:
Army—T
Navy—*MC*

Preparing Activity:
Army—Ordnance Corps

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